



PARASMANI

tubes you can rely on

Engineered for Every Breath.

**MEDICAL GRADE COPPER TUBES
MANUFACTURED TO BS EN 13348**



Internally Degreased



Residual Carbon Controlled

Copper Tubes
by Parasmani





Parasmani

is a leading manufacturer of high-quality copper tubes, serving medical gas, HVAC, refrigeration, plumbing, fuel gas and industrial applications across India. With an annual production capacity exceeding 3,600 MT, Parasmani is recognised for consistent quality and technical precision — supplying hospitals, MGPS turnkey contractors, pharmaceutical facilities and government healthcare projects through a strong Pan-India network and strategic stock points.

Medical-grade copper tube manufacturing is a specialised discipline. What happens inside the bore matters more than anything visible outside it, and the manufacturing route, the degreasing stage and the testing regime are built around that single requirement.

Medical-grade copper tubes are manufactured in strict compliance with: BS EN 13348, ASTM B819

Our Brand



Parasmani is our flagship brand of premium copper tube solutions, trusted across HVAC/R, plumbing, industrial, fuel gas, and medical gas pipeline applications. Under the **Parasmani MGPS** range, our Medical Grade Copper Tubes are engineered to deliver exceptional quality, cleanliness, and reliability for critical healthcare infrastructure.





Pure at the Core

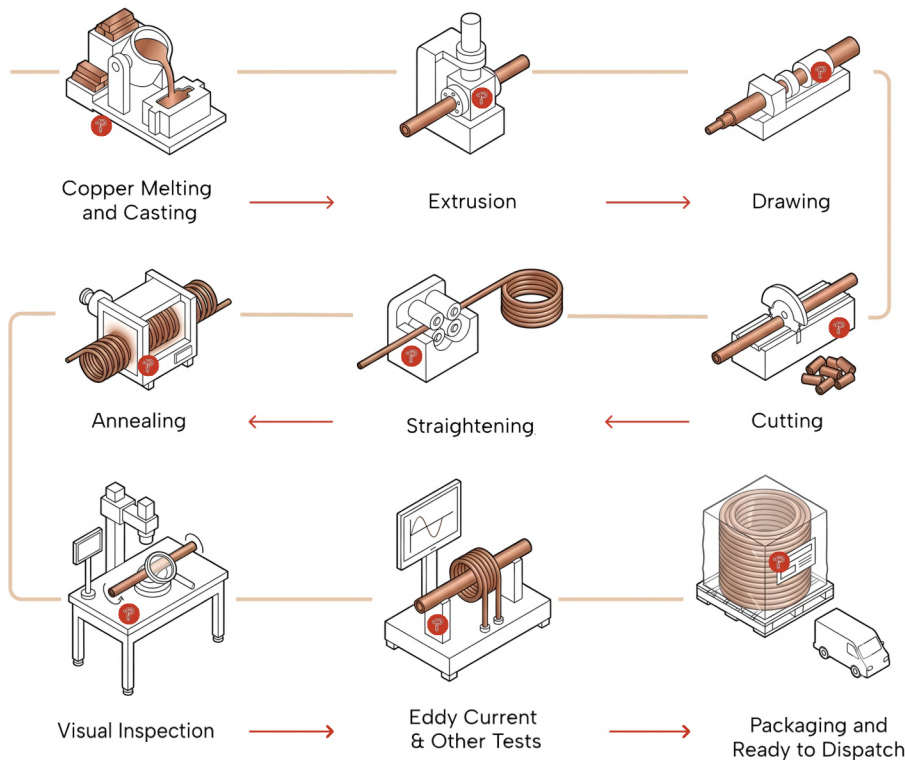
Manufactured from C12200 DHP (phosphorus-deoxidized copper) at 99.90% purity. Superior thermal conductivity, corrosion resistance, and mechanical strength throughout its full cross-section.

Material Grade	C12200 (DHP) / CW024A
Copper Content	Cu 99.90% minimum (including silver)
Phosphorus (deoxidizer)	0.015–0.040%
Temper	Half Hard (R250) standard Hard (R290) on request
Total Impurities	Maximum 0.060% (excluding phosphorus and silver)
Copper Density	8.94 g/cm ³
Copper Melting Point	1084.6 °C
Elongation	30% minimum
Vickers hardness, HV 5 (indicative)	75 to 100
Tensile Strength	250 MPa minimum (R250)





Manufacturing Process



9 - Step Quality Assurance Process

- 01 Chemical Composition Analysis
- 02 Hydrostatic Pressure Test
- 03 Eddy Current Test
- 04 Freedom from Defects
- 05 Pneumatic Pressure Test
- 06 Drift Expanding Test
- 07 Flattening Test
- 08 Hydrogen Embrittlement Test
- 09 Internal Cleanliness Test

Compliance to Standard

Medical gas copper tubes are governed by stricter standards than any other application. Manufactured to BS EN 13348 as the primary medical gas standard and meeting relevant ASTM requirements, every Parasmani tube is internally degreased after the final draw pass, thoroughly dried, and securely end-capped before leaving the factory. Internal cleanliness is verified to the BS EN 13348 limit of 0.20 mg/dm² total residual carbon using a carbon determinator in accordance with the reference method specified in EN 723.

BS EN 13348

Medical Gases & Vacuum

ASTM B819

Medical Gas Tube





Dimensional Matrix as per BS EN 13348

Nominal outside diameters and wall thicknesses as per BS EN 13348, in millimetres.

R – European recommended dimension

X – Other European dimension

Nominal OD (mm)	Nominal Wall Thickness (mm)							Typical Use
	0.7	0.8	0.9	1.0	1.2	1.5	2.0	
6	-	-	-	X	-	-	-	Instrument lines
8	-	R	-	R	-	-	-	Small branch, instrument drops
10	-	R	-	R	-	-	-	Branch lines
12	-	X	-	R	-	-	-	Branch lines, bedhead
14	-	-	-	X	-	-	-	Branch lines
15	R	-	-	R	X	-	-	Ward distribution
16	-	-	-	X	-	-	-	Branch lines
18	-	-	-	R	X	-	-	Sub-main distribution
22	-	-	R	R	X	R	-	Zone valve to ward headers
28	-	-	R	R	X	R	-	Floor headers, OT supply
35	-	-	-	X	R	R	X	Building mains
42	-	-	-	X	R	R	X	Rising mains, plant room
54	-	-	-	X	R	R	R	Main distribution headers
64	-	-	-	-	-	-	R	Large hospital mains
66.7	-	-	-	-	R	-	R	Large hospital mains
76.1	-	-	-	-	-	R	R	Multi-building distribution
88.9	-	-	-	-	-	-	R	Hospital campus main
104	-	-	-	-	-	-	X	Central plant output
108	-	-	-	-	-	R	-	Central plant output

OD 70 and 80 mm, and 108 × 2.5 mm, available on request.

Custom wall thicknesses and cut-to-length available for project requirements.



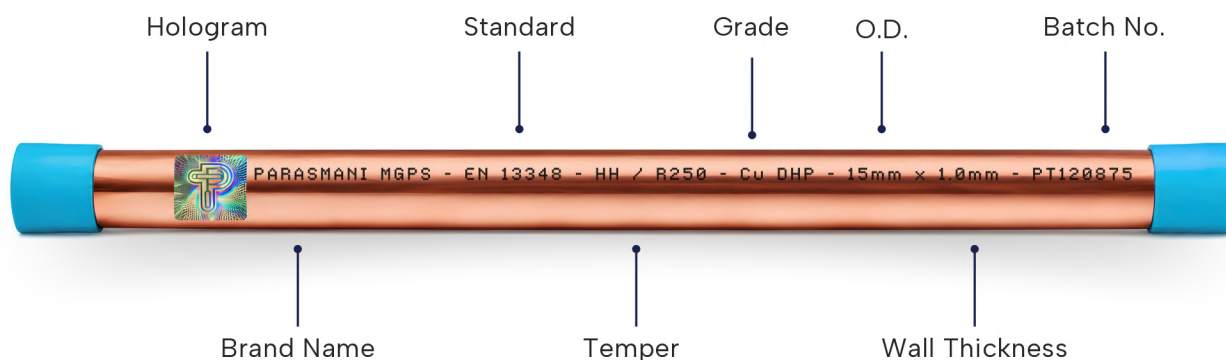
Dimensions, Tolerances & Packing

PARASMANI MGPS STRAIGHT COPPER TUBES | STANDARD PIPE LENGTH: 3 MTR

O.D. (mm)	W.T. (mm)	O.D. Tolerance	W.T. Tolerance	Weight Per Mtr. (kg)	Pipes Per Bundle	Weight Per Bundle
10	0.9	±0.09	±0.09	0.231	50	34.65
12	0.7	±0.09	±0.07	0.223	50	33.45
12	1.0	±0.09	±0.10	0.309	40	37.08
15	0.7	±0.09	±0.07	0.282	40	33.84
15	0.9	±0.09	±0.09	0.357	40	42.84
15	1.0	±0.09	±0.10	0.394	35	41.37
22	0.9	±0.10	±0.09	0.534	25	40.05
22	1.0	±0.10	±0.10	0.590	20	35.40
28	0.9	±0.10	±0.09	0.686	20	41.16
28	1.0	±0.10	±0.10	0.759	15	34.16
35	1.2	±0.11	±0.12	1.140	10	34.20
42	1.2	±0.11	±0.12	1.376	10	41.28
54	1.2	±0.11	±0.12	1.780	8	42.72
76.1	1.5	±0.15	±0.225	3.143	4	37.72
108	1.5	±0.30	±0.225	4.487	3	40.38

Custom wall thicknesses and additional sizes available on request — contact sales.

Tube Marking Scheme





Why Half Hard Matters in MGPS

Half Hard (R250) is the recommended temper under BS EN 13348 for medical gas pipelines and is therefore the standard temper supplied by Parasmani. It resists stress cracking, absorbs vibration, and accommodates thermal movement for long-term reliability.



Property	Half Hard R250	Hard Drawn R290
Long-term stress cracking	Resistant	More prone to stress cracking
Vibration and hammering	Absorbed by the tube	Transmitted to joints
Field bending / uneven surfaces	Accommodates movement	Higher risk of cracking
Thermal expansion & contraction	Absorbed by inherent ductility	Lower tolerance to movement
Seismic movement	Tolerant	Lower tolerance

Jointing and Brazing

Use end-feed capillary fittings to BS EN 1254-1:2021, factory degreased for oxygen service and supplied individually sealed in protective packaging. Insert the tube fully to the shoulder stop. Braze with CP 104 copper-silver-phosphorus filler to BS EN ISO 17672 — no flux is required on copper-to-copper joints. Indicative minimum brazing temperature 710 °C. Penetration must reach three times the wall thickness or 3 mm, whichever is greater.

Nitrogen shielding. Oxygen-free nitrogen (OFN) must flow through the pipework throughout brazing to prevent oxide scale forming on the bore. Supply it to the assembled, un-brazed line via a regulator and flow controller; flush first to clear air, then maintain a continuous or intermittent flow. Purge after brazing to remove shield gas and particulate. This preserves the internal cleanliness achieved during manufacturing — the condition an oxygen service depends on.

Ensure the torch reaches brazing temperature across the full joint. Oxy-acetylene or oxygen-fed LPG is recommended; air-fed LPG may not reach temperature on larger sizes. Avoid prolonged heating, which can damage the tube and fitting.

In confined spaces, ventilate to prevent nitrogen build-up and monitor ambient oxygen content.



Copper Fittings

Medical gas pipeline fittings are end-feed type, manufactured from the same grade of copper as the tube and in accordance with BS EN 1254-1. Fittings are factory degreased, suitable for oxygen service and supplied individually sealed in protective polythene, preserving the cleanliness of the tube until installation.

PARASMANI COPPER FITTINGS | PRODUCT RANGE & SPECIFICATIONS | BS EN 1254-1

Fitting Type	Size Range – O.D. (mm)	≈ Size Range (inches)	Wall Thickness (mm)	≈ Wall Thickness (SWG)
90° Elbow C × C	6 mm – 108 mm	3/16" – 4"	0.6 mm – 2.5 mm	24 SWG – 12 SWG
Coupling C × C	6 mm – 108 mm	3/16" – 4"	0.6 mm – 2.5 mm	24 SWG – 12 SWG
Reducing Coupling C × C	8 × 6 mm – 108 x 76 mm	1/4" × 3/16" – 4" × 3"	0.6 mm – 2.5 mm	24 SWG – 12 SWG
Equal Tee C × C × C	6 mm – 108 mm	3/16" – 4"	0.6 mm – 2.5 mm	24 SWG – 12 SWG
Reducing Tee C × C × C	8 × 6 mm – 108 x 76 mm	1/4" × 3/16" – 4" × 3"	0.6 mm – 2.5 mm	24 SWG – 12 SWG



Contact Us

Get in touch today for pricing, technical specifications, and project enquiries.



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